

Demonstration of Proposed Features

Date	01 July 2026
Team ID	XXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Data Preprocessing & Feature Scaling	Handling missing values, outlier detection, encoding, and feature scaling on the flood dataset	Implemented	Yes	Verified on Kaggle flood dataset
2	Multi-Model Classification	Training and comparing Decision Tree, Random Forest, KNN, and XGBoost classifiers	Implemented	Yes	All four models trained and evaluated
3	Model Evaluation & Selection	Evaluating models via confusion matrix, classification report, and accuracy score to select the best model	Implemented	Yes	XGBoost selected at 96.55% accuracy
4	Model Persistence	Saving the trained model and scaler for deployment	Implemented	Yes	Saved as floods.save
5	Flask Web Application	Home page, prediction input page, and result pages built with Flask/Jinja2	Implemented	Yes	Demonstrated end-to-end on local server
6	Real-time Flood Risk Prediction	User enters rainfall/weather values and instantly views flood chance result	Implemented	Yes	Live prediction demoed with sample inputs

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%